



P25

Agent Task Markets: VCG, Private Information, and Settlement Neutrality

Agent 任务市场：VCG 私有信息与结算中立性

TwinsEarth · 智能体宇宙论文系列



Agent Task Markets: VCG, Private Information, and Settlement Neutrality

Working Title (EN): Agent Task Markets: VCG, Private Information, and the Design Goals of Settlement Neutrality

Shard D • Market and Architecture paper P25 • the open-market extension of v7.5.0 P11 "Settlement conservation of the internal task market" • cross-referenced with P17 (least privilege), P10 (TransferBundle) • UDOS Reasoning Engine v7.6.0 • Fang Wenxin • 2026-09-21

One sentence first: when task allocation among Agents moves from "bookkeeping inside one closed system" to "a cross-system open market with private information," the old "cost-performance award" is no longer enough—the real difficulty is not "who wins," but "**how to make Agents willing to state their true cost, while keeping the ledger from creating or losing money out of thin air during cross-system settlement**"; this paper splits this problem into three lines: VCG-style incentives (making truth-telling pay),

private-information mechanism design (making bids carry strategy notes), and "settlement neutrality," a UDOS-authored engineering design goal.

Evidence caliber (stated only once in the whole paper): this paper is a **design survey + proposal**, containing no new UDOS measured numbers. All three external dialogue objects were verified online 2026-09-21: AgentLance (arXiv:2608.23867, verified), Agent Exchange (arXiv:2507.03904, verified), ISONOMIA (**correctly recorded as Zenodo records/21338480**, caution; its "ergs mutual-trust credit unit, bicameral governance" are true, while numbers such as "45,000 whole-economy simulations / 300 parameter points" were not verified, all [RESULT NEEDED]). Caution: the "five attributes of settlement neutrality" in the user's clue **has no traceable original source** (ledger failed), this paper does not cite it as an existing framework, but rewrites it as a **UDOS P11/P25-authored design goal**, marking [CITATION NEEDED] where academic backing is needed. UDOS old numbers only point back to v7.5.0 P11 (settlement conservation invariant, 10 unit tests verified, matrix clean/stressed both conserved=true).

Abstract

Multi-agent orchestration is moving from "internal dispatch" to "market dispatch." v7.5.0's P11 already wrote "the books balance" of a closed internal market as a settlement conservation invariant guaranteed by code structure ($\text{balance_sum} = \text{total_paid} - \text{slashed}$, $\text{total_paid} \leq \text{total_budget}$). But once the view extends to a cross-system open Agent market with private cost information, three new problems follow: first, bidders **know their true cost while buyers do not**, "cost-performance award" is no longer transparent under private information; second, VCG's elegant "truth-telling dominant" mechanism is engineering-expensive, collusion-prone, and does not satisfy budget balance (Ausubel–Milgrom long pointed out its "theoretically elegant, practically lonely"); third, during cross-system settlement, the ledger bottom line "money is not created or destroyed out of thin air" cannot be guaranteed only by a single system's internal invariants.

This paper surveys along three lines and gives UDOS's extension design: **(1) VCG and private information**—AgentLance (arXiv:2608.23867) does decentralized orchestration with "private-cost bidding + self-maintained strategy notes + VCG-style payment," this paper thereby distinguishes "closed cost-performance award" from "open incentive-compatible award"; **(2) auctions and liquidity layering**—Agent Exchange (arXiv:2507.03904) borrows from real-time ad bidding, using multi-attribute auctions to degrade to greedy/random allocation when liquidity is insufficient; **(3) credit units and governance**—ISONOMIA (Zenodo 21338480) uses "ergs," a mutual-trust bookkeeping unit generated only at verified settlement with net zero, and bicameral governance, answering "what does an open market use to keep accounts, who governs."

On top of the three, this paper explicitly makes "settlement neutrality" a **UDOS-authored design goal** (rather than citing an existing framework), and honestly states it currently lacks academic backing [CITATION NEEDED]. All new designs in this paper are **design proposals • to implement**, not claiming to have been measured in an open market.

Keywords: Agent task market; VCG; private information; strategy notes; settlement conservation; settlement neutrality; internal market; open market; engineering mechanism design

Structured Abstract (background problem → method/argument → evidence → contribution)

- **Background problem:** P11's closed internal market solved "the books balance," but the open Agent market introduces three problems P11 did not face: private cost information, cross-system settlement, real governance.
 - **Method/argument:** splitting open-market dispatch into three orthogonal lines—the incentive line (VCG/truth-telling), the allocation line (auctions/liquidity degradation), the bookkeeping-governance line (credit units/governance structure); and raising "settlement neutrality" as a UDOS-authored engineering design goal.
 - **Evidence clues:** (1) AgentLance's private cost + strategy notes + VCG-style payment (verified, paper reports it beats single models and centralized allocation); (2) Agent Exchange's multi-attribute auctions and liquidity degradation (verified); (3) ISONOMIA's ergs credit unit and bicameral governance (caution, simulation numbers not verified); (4) P11's verified settlement conservation invariant as the closed baseline.
 - **Contribution:** (1) layering "closed cost-performance award" and "open incentive-compatible award"; (2) giving a design map of the open market's three orthogonal lines; (3) clarifying "settlement neutrality" is a UDOS-authored goal and marking its evidence gap; (4) marking all new designs [RESULT NEEDED], not exaggerating.
-

1 Introduction: From "Dispatch" to "Market," the Problem Changed

P11 faces a **closed internal market**: tasks first priced, Agents bid, awarded by "quality threshold + cost-performance + load fallback," paid only to the unique completer after QA acceptance, Byzantine deposits slashed. It wrote "the books balance" as a conservation invariant, and proved this conservation is **a theorem guaranteed by the settlement code structure**, not luck.

But "internal dispatch" and "market dispatch" differ essentially. In a real, cross-organizational Agent ecosystem, three things P11 did not face appear:

1. **Private cost information:** bidders know their true cost (they know their compute, they know their private data allowance), while buyers and other Agents do not. P11's Bid "cost" is self-reported by the bidder, enough in a closed system; in an open system, **self-reported cost can lie**.
2. **From "award" to "allocation":** in a closed system the Agent pool is fixed, award rules simple; in an open system one must face liquidity—sometimes many bidders, sometimes not one, rules must degrade reasonably under different liquidity.
3. **What cross-system uses to keep accounts, who governs:** P11's ledger is a single system's internal bookkeeping unit, invalid outside that system. Cross-system settlement needs a "credit unit" and a governance structure of "who has the final say."

This paper pushes P11's closed ledger one step in each of these three open directions. But before pushing, stay calm: **VCG, the prettiest mechanism in textbooks, is precisely the hardest to land in engineering**. P11 already cited Ausubel and Milgrom's sober judgment—VCG is "theoretically elegant, practically lonely": compute-expensive, collusion-prone, not budget-balanced, rarely deployed at scale in real market design. This paper's orientation is therefore not "move VCG over," but "see clearly what problem VCG wants to solve, then approximate it within what engineering can bear."

2 The Incentive Line: VCG, Private Cost, and Strategy Notes



Figure 1 Left: closed internal market—cost visible, cost-performance award, conservation guaranteed by code structure (P11 verified); right: open market—cost private, needs incentive compatibility making truth-telling pay, needs cross-system bookkeeping. Conceptual illustration, not measured data.

2.1 What Problem VCG Wants to Solve

VCG (Vickrey–Clarke–Groves)'s core idea in one sentence: **make each bidder pay "the externality its presence imposes on others,"** so truth-telling becomes the dominant strategy. Under ideal assumptions it simultaneously achieves incentive compatibility (truth-telling optimal) and Pareto efficiency. The problem is engineering: it needs public preference aggregation, combinatorial-explosion compute, and does not satisfy budget balance.

P11's choice is stepping back—not pursuing "truth-telling dominant," but pursuing "the books balance": using "pay only on acceptance, pay only the unique completer, Byzantine necessarily slashed" to suppress duplicate labor and false reporting, exchanging VCG's elegance for 300 lines of testable conservation. This paper continues this orientation.

2.2 AgentLance: Private Cost + Strategy Notes

AgentLance (Markets, Not Planners: Decentralized Orchestration of LLM Agents with Private Information, arXiv:2608.23867, verified) faces precisely the "private cost" point. Its mechanism (paper report):

- Agents bid with **private cost**—they know their cost but do not disclose it to others;
- each Agent self-maintains **strategy notes**, recording how it historically bid and the outcomes;
- the allocator picks winners based on bids and **public reputation records**;
- adopting **VCG-style payment**—making bidders' bids reflect true cost.

The paper reports it beats single models, centralized allocation and market baselines on math reasoning, code generation, knowledge-intensive QA and Agent tasks (paper report • not independently reviewed).

The borrowing for UDOS is not "copying the VCG payment formula," but two structural judgments:

1. **Private information is a first-class variable.** P11's Bid.cost is self-reported and closed; if UDOS wants an open market, it must directly answer "whether self-reported cost is trustworthy." AgentLance's answer is: **not assuming it trustworthy, but designing payment rules making truth-telling beneficial to oneself** (VCG-style), and using **public reputation** as external correction.
2. **"Strategy notes" are a self-governing memory worth borrowing.** Bidders do not bid from zero each time, but bid with notes of historical bidding experience. This is spiritually consistent with v7.6.0 P18's "BRS+DRS memory flywheel"—both compress "past experience" into reusable internal memory, rather than deciding from scratch each time.

2.3 UDOS Layering: When to Use Cost-Performance, When to Approximate VCG

This paper proposes layering UDOS's award rules by "information openness":

- **Closed, same system, cost approximately visible** → continue P11's "quality threshold + cost-performance + load fallback," conservation guaranteed by code structure (verified).
- **Open, cross-system, cost private** → adopt the approximate mechanism of "VCG-style payment + public reputation correction," the goal being to make **truth-telling more pay than lying**, rather than strictly dominant.

Why "approximate" rather than "strict VCG"? Because P11 already made clear: strict VCG is compute-expensive, collusion-prone, not budget-balanced. UDOS's orientation is engineering auditability first—approximate incentive compatibility where possible, but never pretend to have proven strict dominance.

3 The Allocation Line: Auctions, Liquidity, and Allocation Degradation

3.1 Agent Exchange: An Ad-Bidding-Style Agent Market

Agent Exchange (Agent Exchange: Shaping the Future of AI Agent Economics, arXiv:2507.03904, verified) borrows real-time ad bidding (RTB) to design Agent task allocation: treating tasks as "impression slots," Agents as "advertisers," using **multi-attribute auctions** (looking not only at price but quality, capability, latency) for allocation. Its reported mechanisms (paper report) include:

- an RTB-style auction platform (component divisions such as USP/ASP/Agent Hubs/DMP);
- multi-attribute auctions + generalized second price;
- **degradation when liquidity is insufficient**—when bidders are not enough, degrading to greedy or random allocation, rather than stuck at "must auction."

The borrowing for UDOS is "**liquidity awareness**." P11's award rules assume bidders are sufficient; open markets often lack enough bidders. A good allocation rule should be **adaptive degradation**: precise auctions when many, degrading to "nearest dispatch" or "random fallback" when few, rather than shutting down because bidders cannot be assembled.

3.2 From "Who Wins" to "How to Degrade"

This section's real argument is: the design difficulty of open-market allocation rules lies not in "how to pick the optimum when bidders are sufficient" (auction theory already solved this), but in **"how to degrade gracefully when bidders are insufficient, information incomplete, transient faults occur."** This is spiritually fully consistent with UDOS P3's circuit breaker, P2's view change—a good system does not run beautifully under ideal conditions, but does not crash when ideal conditions break.

4 The Bookkeeping and Governance Line: ergs Credit Unit and Bicameral Governance

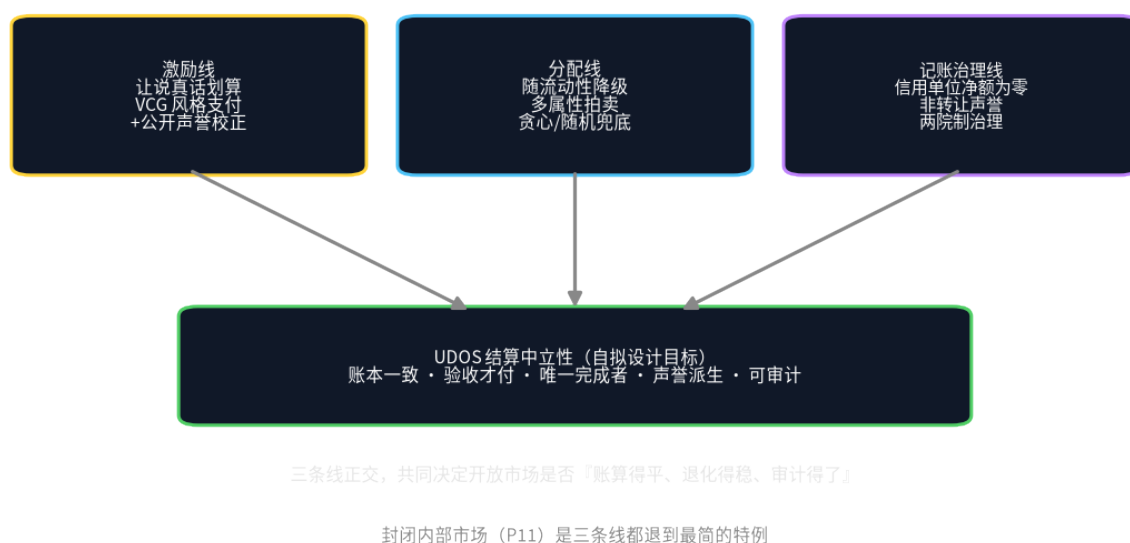


Figure 2 The three orthogonal design lines of the open Agent market—incentives (making truth-telling pay), allocation (degrading with liquidity), bookkeeping governance (credit unit + governance structure). P11 closed conservation is its special case. Conceptual illustration, not measured data.

4.1 ISONOMIA: The Net-Zero Mutual-Trust Credit Unit

ISONOMIA (ISONOMIA: A Constitutional Design for Autonomous Agent Labor Markets, **correctly recorded as Zenodo records/21338480**, caution) faces "what an open market uses to keep accounts." Its core design (paper report):

- **ergs**: a mutual-trust bookkeeping unit, **generated only as debit-credit pairs at verified settlement, net zero**—i.e. not "printing money out of thin air," but "one debit, one credit, netting to zero";
- **non-transferable reputation**: reputation follows behavior, cannot be resold like a token;
- **bicameral governance**: one chamber voting weighted by accumulated prestige, one chamber voting equal by lot—using the dual track of "capability chamber + equal-rights chamber" to avoid governance being monopolized by a few strong players.

The borrowing for UDOS is very direct, and strikingly echoes P11's conservation: P11 requires "money not created or destroyed out of thin air" ($\text{balance_sum} = \text{total_paid} - \text{slashed}$), ISONOMIA's ergs require "each entry generated in pairs, net zero." **This is the same conservation intuition, one implemented inside a single system with code invariants, one implemented in an open economy with bookkeeping-unit design.** This shows "the books balance" is not UDOS's private good, but a common bottom line across scales of markets.

Caution, honestly marked: ISONOMIA's simulation numbers such as "45,000 whole-economy simulations / 300 parameter points $\times$ 50 seeds / 300-300 pass termination criterion" **were not verified in this check**, this paper does not cite them as facts, all [RESULT NEEDED]. This paper only adopts the two structural facts of its **bookkeeping-unit design and governance structure.**

4.2 Governance: Why an Open Market Needs "Two Chambers"

A closed system's governance is UDOS itself having the final say (P3's circuit breaker, P12's evidence grading). An open market has no single authority, so governance itself becomes a designed object. ISONOMIA's bicameral governance gives a simple but deep tradeoff: **purely weighting by prestige** lets governance be monopolized by a few strong players (the strong always win); **purely equal by lot** lets governance be hostage to the average level. The bicameral compromise is—one side listening to "those who did more, did better," one side listening to "one person one vote."

UDOS does not claim to copy bicameral governance, but incorporates into the open-market design map the judgment that **"the governance structure itself needs to be designed, rather than an authority existing by default."**

5 Settlement Neutrality: A UDOS-Authored Design Goal (with evidence gap statement)

5.1 Why a Separate Section, and Must First State Its Source

The user's clue once contained the claim of "five attributes of settlement neutrality" (escrow determinism, ledger consistency, reputation equivalence, dispute reciprocity, auditability). **After online verification, no original paper or Zenodo record of this claim was found** (ledger failed). Therefore this paper **does not cite it as an existing academic framework**, but explicitly rewrites it as a **UDOS P11/P25-authored engineering design goal**: when settlement moves from "a single closed system" to "a cross-system open market," UDOS wants the settlement protocol to satisfy the following engineering properties. The five below are **UDOS-authored**, marking [CITATION NEEDED] where academic backing is needed:

1. **Ledger consistency (UDOS-authored)**: from whichever Agent's perspective, the same settlement points to the same set of facts on the ledger; conservation such as $\text{balance_sum} = \text{total_paid} - \text{slashed}$ still holds after cross-system netting.
2. **Acceptance-dependent settlement (UDOS-authored)**: payment occurs only after work is accepted, no positive settlement without acceptance (consistent with P11 "pay only on acceptance").
3. **Unique-completer payment (UDOS-authored)**: the same task settles only with one accepted completer, duplicate submissions produce no duplicate payment (P11 verified).
4. **Reputation derived from behavior (UDOS-authored)**: reputation is derived from real settlement outcomes, cannot be bought or transferred (borrowing ISONOMIA's non-transferable reputation).

5. **Auditability (UDOS-authored):** each settlement can be traced along the trace back to acceptance evidence and contract fields (P10's hash-chain Trace).

These five are not "theorems academia has proven," but engineering expectations UDOS **induces** from P11's conservation, P10's auditability, ISONOMIA's credit units. Writing them as a list is to give later implementation a target, not to claim it is already consensus.

5.2 Demarcation from the "Token Economy / Agent Currency" Narrative

P11 already made clear: UDOS's market is an **internal bookkeeping unit**, not linked to any token/financialization, no real pricing game, no Sybil defense. P25 extends the view to the open market, but **this demarcation is unchanged**: this paper discusses "mechanism design engineering," not "issuing tokens." Any industry narrative of "issuing currency to Agents, letting the market self-regulate" needs Sybil defense, price discovery, external trust anchors, far beyond this paper's scope, this paper not mutually endorsing it.

6 Drawbacks / Failure Boundaries / Falsifiable Conditions

Its drawbacks and limitations:

1. **VCG's "elegance trap."** The easiest error is "since VCG makes truth-telling optimal, then fully adopt VCG." Ausubel–Milgrom long warned: VCG is compute-expensive, collusion-prone, not budget-balanced. If UDOS hard-adopts strict VCG in the open market, it falls into the trap of "theoretically perfect, engineering unable to run." This paper insists on "approximate VCG + reputation correction," not pursuing strict dominance.
2. **Private information ≠ designable.** AgentLance's mechanism works in its experiments, but "private cost" in a real Agent ecosystem may be manipulated more complexly (Agents can falsely report their own strategy notes). VCG-style payment suppresses "bid lying," not "the strategy notes themselves being polluted."
3. **ISONOMIA's numbers cannot be trusted as facts.** Its ergs/bicameral governance is structural design, but its "45,000 simulations all pass" is an unverified number, this paper not citing. Whether a bookkeeping-unit design can really stabilize in an open economy is itself an open question.
4. **Settlement neutrality is an authored goal, not a theorem.** Section 5's five are UDOS's expectations, no academic backing, nor proven "satisfying these five means safe." Taking them as design targets is fine, as safety guarantees is not.
5. **Open market ≠ free market.** This paper still discusses a controlled open market that is "auditable, contracted, accepted," not "unregulated free competition." Once acceptance and contracts are removed, all properties fail.

Falsifiable conditions:

- **F1:** if in open-market simulation, VCG-style payment + reputation correction does **not** raise the proportion of "truth-telling bids" (relative to the cost-performance award baseline), then the judgment "private cost must be treated with incentive compatibility" is wrong in the Agent scenario, should fall back to cost-performance award.
- **F2:** if after cross-system netting, the conservation invariant ($\text{balance_sum} = \text{total_paid} - \text{slashed}$) is frequently violated in real settlement, then "settlement neutrality" as an engineering goal is unrealizable, UDOS should shrink back to the closed internal market.

- **F3:** if liquidity-adaptive degradation (auctions when many, fallback when few) in simulation is instead worse than fixed rules due to "degradation too coarse," then "liquidity awareness" is over-design.
-

7 Conclusion and Honest Boundary

P11 answered "how a closed internal market balances the books," P25 pushes the view to "how an open Agent market dispatches, keeps accounts, governs." Three orthogonal lines—incentives (VCG-style making truth-telling pay), allocation (adaptive degradation with liquidity), bookkeeping governance (ergs credit unit + bicameral governance)—together form a design map. On it, UDOS authors "settlement neutrality" as an engineering goal, and honestly states it lacks academic backing [CITATION NEEDED].

All new designs in this paper are **design proposals • to implement:** whether VCG-style payment really raises truth-telling proportion in the open Agent market, whether cross-system conservation holds, whether liquidity degradation is better—all [RESULT NEEDED]. UDOS does not pursue making the market "the theoretically prettiest auction," but an engineering-reproducible market that "balances the books, degrades stably, is auditable." This is P25's increment for v7.6.0.

References

External dialogue objects (existence verified online 2026-09-21; numbers are paper reports or record actual values)

1. AgentLance: Markets, Not Planners: Decentralized Orchestration of LLM Agents with Private Information. arXiv:2608.23867 (private-cost bidding, strategy notes, VCG-style payment; its beating single models/centralized allocation is a paper report).
2. Agent Exchange: Shaping the Future of AI Agent Economics. arXiv:2507.03904 (RTB-style multi-attribute auctions, generalized second price, greedy/random degradation when liquidity insufficient).
3. ISONOMIA: A Constitutional Design for Autonomous Agent Labor Markets. Zenodo records/**21338480** (caution, correctly recorded; ergs mutual-trust credit unit net zero, non-transferable reputation, bicameral governance; numbers such as "45,000 simulations" not verified, [RESULT NEEDED]).

Classic background

1. Vickrey, W. (1961). Counterspeculation, Auctions, and Competitive Sealed Tenders. Journal of Finance.
2. Ausubel, L. M., & Milgrom, P. (2006). The Lovely but Lonely Vickrey Auction. (the classic warning of VCG "theoretically elegant, practically lonely"; relayed from P11).

UDOS paper volumes

1. UDOS v7.5.0: P11 "Settlement conservation and Byzantine slashing mechanism design of the internal task market" (conservation invariant, 10 unit tests verified, clean/stressed both conserved=true), P10 "TransferBundle handoff contract" (hash-chain Trace, 26 tests verified).
 2. UDOS v7.6.0: P17 "Least-privilege orchestration" (winners execute only within scope), P18 "Broad-deep two-stage memory flywheel" (strategy notes and memory flywheel isomorphic), P34 "Literature evidence ledger."
-

Evidence Discipline and Reproduction Notes

- This paper is a **design survey + proposal**, no new UDOS measurements; VCG-style payment, liquidity degradation, the five settlement neutrality items are all design proposals, benefit positions [RESULT NEEDED] .
- The "five attributes of settlement neutrality" **has no traceable original source** (ledger failed), this paper does not cite it as an existing framework, rewriting it as a UDOS-authored design goal and marking [CITATION NEEDED] .
- ISONOMIA is correctly recorded as Zenodo 21338480 (not the user-given 21287289); its simulation numbers were not verified, all not written as facts.
- AgentLance/Agent Exchange's "beating baselines" is a paper report, not independently reviewed; not mutually endorsed with any "Agent token economy" industry narrative.